

ADC HW-1

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Exercise 3.3 #

$$\|u\|_2^2 = \langle u, u \rangle = \int_0^1 (t+2)^2 \cdot dt$$

$$= \frac{(t+2)^3}{3} \Big|_0^1$$

$$= \frac{3^3}{3} - \frac{2^3}{3} = \frac{27-8}{3}$$

$$= \frac{19}{3}$$

$$\|v\|_2^2 = \int_0^1 \underbrace{(t^2 - 2t)}_a \underbrace{+ 3}_b^2 \cdot dt$$

$$= \int_0^1 (t^2 - 2t)^2 - 2(t^2 - 2t)(3) + (3)^2 \cdot dt$$

$$= \int_0^1 t^4 - 4t^3 + 4t^2 - 6t^2 + 12t + 9 \cdot dt$$

$$= \int_0^1 t^4 - 4t^3 - 2t^2 + 12t + 9 \cdot dt$$

$$= \frac{t^5}{5} \Big|_0^1 - t^4 \Big|_0^1 - \frac{2t^3}{3} \Big|_0^1 + 6t^2 \Big|_0^1 + 9t \Big|_0^1$$

$$= \frac{1}{5} - 1 - \frac{2}{3} + 6 + 9 = \frac{203}{15}$$

$$\langle u, v \rangle = \int_0^1 (t+2)(t^2-2t-3) \cdot dt.$$

$$= \int_0^1 t^3 - \cancel{2t^2} - 3t + \cancel{2t^2} - 4t - 6 \cdot dt.$$

$$= \int_0^1 t^3 - 7t - 6 \cdot dt.$$

$$= \left. \frac{t^4}{4} \right|_0^1 - \left. \frac{7t^2}{2} \right|_0^1 - \left. 6t \right|_0^1$$

$$= \frac{1}{4} - \frac{7}{2} - 6 = \boxed{\frac{-74}{8}}$$

### Exercise 3.2 #

$$\langle \vec{x}, \vec{y} \rangle = \int_{-\infty}^{+\infty} x(-t) \cdot y(-t) \cdot dt.$$

$$= \int_{+\infty}^{-\infty} -x(v) y(v) \cdot dv$$

$$-t = v$$

$$t = -v$$

$$= \int_{-\infty}^{+\infty} x(v) y(v) \cdot dv.$$

$$dt = -dv.$$

### Exercise 3.3 #

$$(i) \quad E[x(\cdot - t_0)] = \int_{-\infty}^{\infty} |x(t - t_0)|^2 \cdot dt.$$

$$v = t - t_0$$

$$dv = dt$$

$$E[x(\cdot - t_0)] = \int_{-\infty}^{\infty} |x(u)|^2 \cdot du = \boxed{E[x]}$$

(ii)

$$E[x(-\cdot)] = \int_{-\infty}^{\infty} |x(-t)|^2 \cdot dt$$

$$u = -t, \text{ so}$$

$$du = -dt,$$

$$E[x(-\cdot)] = \int_{\infty}^{-\infty} |x(u)|^2 \cdot du$$

$$= \int_{-\infty}^{\infty} |x(u)|^2 \cdot du = \boxed{E[x]}$$

(iii) All three energies are equal to each other.

$$E[x(t)] = E[x(t - t_0)] = E[x(-t)].$$

Exercise 3.4 #

$$\Rightarrow \langle u, 3u + v + iw \rangle + \langle v, 3u + v + iw \rangle$$

$$\Rightarrow \langle u, 3u \rangle + \langle u, v \rangle + \langle u, iw \rangle + 3\langle v, u \rangle + \langle v, v \rangle + \langle v, iw \rangle.$$

$$\Rightarrow 3\|u\|_2^2 + \|v\|_2^2 + \langle u, v \rangle + 3\langle u, v \rangle^* - i\langle u, w \rangle - \langle v, w \rangle$$



## Exercise 3.5 #

$$\begin{aligned}
 (i) \quad \langle u-v, u+v \rangle &= \langle u, u+v \rangle - \langle v, u+v \rangle \\
 &= \langle u, u \rangle + \langle u, v \rangle - \langle v, u \rangle + \langle v, v \rangle \\
 &= \|u\|_2^2 - \|v\|_2^2
 \end{aligned}$$

(ii) No, check it using  $u = a+bi$ ,  $v = c+di$

## Exercise 3.6 #

if  $\mu(\{t: x(t) \neq 0\}) = 0$ , means  $0 \equiv x$ .

Let  $u(t)$  be integrable  $\Rightarrow \langle u, 1 \rangle$  is defined.

$$\int |u(t)| \cdot 1 \, dt = 0 \Leftrightarrow \langle u, 1 \rangle = 0$$

$$\Downarrow$$

$$u \equiv 0$$

If  $\|u\|_2^2 < \infty$  i.e.  $u$  is an energy limited function then we need a different strategy.

$$\text{Suppose } u \equiv 0 \Rightarrow \mu(\{|u(t)|^2 \neq 0\}) = 0$$

$$\text{because } \underbrace{\{|u(t)|^2 \neq 0\}}_A \Rightarrow \underbrace{\{|u(t)| \neq 0\}}_B.$$

$$t \in A \Rightarrow t \in B.$$

$$A \subseteq B.$$

$$\text{Thus, } \mu(\{ |v(t)|^2 \neq 0 \}) = 0.$$

$$\text{and } |v(t)|^2 = 0.$$

$$\Downarrow$$

$$\int_K |v(t)|^2 \cdot dt = 0$$

$$\Downarrow$$

$$\|v\|_2^2 = 0 \Rightarrow \|v\|_2 = 0$$

$$\Downarrow$$

$$|\langle v, v \rangle| \leq \|v\|_2 \|v\|_2 = 0$$

$$\Downarrow$$

$$\langle v, v \rangle = 0 \quad \forall \quad v$$

$$\text{Also if } \langle v, v \rangle = 0 \quad \forall \quad v \Rightarrow \langle v, v \rangle = 0 \Rightarrow \|v\|_2^2 = 0$$

$$\Rightarrow \int |v(t)|^2 \cdot dt = 0 \Rightarrow \int |v(t)| \cdot dt = 0 \Rightarrow v \equiv 0.$$

### Exercise 3.7 #

$$|\langle v, v \rangle| = \frac{1}{2} (\|v\|_2^2 + \|v\|_2^2) \quad \text{--- (i)}$$

$$|\langle v, v \rangle| \leq \|v\|_2^2 \quad \text{--- (ii)}$$

$$\|v\|_2 \|v\|_2 \leq \frac{1}{2} (\|v\|_2^2 + \|v\|_2^2)$$

$$0 \leq \|v\|_2^2 + \|v\|_2^2 - 2\|v\|_2 \|v\|_2$$

$$= (\|v\|_2 - \|v\|_2)^2$$



Exercise 3.8 #

$$u_1(t) = u(t) \sqrt{h(t)}$$

$$v_1(t) = v(t) \sqrt{h(t)}$$

$$\left| \int_{-\infty}^{\infty} u_1(t) v_1^*(t) \cdot dt \right|^2 \leq \left( \int_{-\infty}^{\infty} |u_1(t)|^2 \cdot dt \right) \left( \int_{-\infty}^{\infty} |v_1(t)|^2 \cdot dt \right)$$

since,  $u_1(t) v_1^*(t) = u(t) \sqrt{h(t)} \cdot (v(t) \sqrt{h(t)})^* = u(t) v^*(t) h(t)$ .

and  $|u_1(t)|^2 = |u(t)|^2 h(t)$ ,  $|v_1(t)|^2 = |v(t)|^2 h(t)$ .

$$\left| \int_{-\infty}^{\infty} u(t) v^*(t) h(t) \cdot dt \right|^2 \leq \left( \int_{-\infty}^{\infty} |u(t)|^2 h(t) \cdot dt \right) \left( \int_{-\infty}^{\infty} |v(t)|^2 h(t) \cdot dt \right)$$

Exercise 3.9 #

$$\begin{aligned} &= |\langle u, v \rangle - \langle u', v' \rangle| \\ &= |\langle u, v \rangle - \langle u', v \rangle + \langle u', v \rangle + \langle u', v' \rangle| \\ &= |\langle u - u', v \rangle + \langle u', v - v' \rangle| \\ &\leq |\langle u - u', v \rangle| + |\langle u', v - v' \rangle| \\ &\leq \|u - u'\|_2 \|v\|_2 + \|u'\|_2 \|v - v'\|_2. \end{aligned}$$

because  $u \equiv u'$

$$u - u' \equiv 0$$

$$(u - u')^2 \equiv 0$$

$$\|u - u'\|_2 = 0.$$

$$\Downarrow$$

0

Exercise 3.10 #

$$(i) \quad E[x] = \int_{-\infty}^{\infty} |x(t)|^2 \cdot dt.$$

$$\int_{-\infty}^{\infty} |x(t)|^2 (1+t^2) \cdot dt = \int_{-\infty}^{\infty} |x(t)|^2 \cdot dt + \int_{-\infty}^{\infty} |t(x(t))|^2 \cdot dt$$

||

$$E[x] + E[tx]$$

$$x \in L^2(\mathbb{R}) \quad , \quad tx \in L^2(\mathbb{R})$$



$$\int_{-\infty}^{\infty} |x(t)|^2 (1+t^2) \cdot dt < \infty.$$

$$(ii) \quad \int |x(t)| \cdot dt = \int |x(t)| \cdot \frac{1}{\sqrt{1+t^2}} \cdot \sqrt{1+t^2} \cdot dt.$$

$$\leq \left( \int_{-\infty}^{+\infty} \frac{1}{1+t^2} \cdot dt \right)^{1/2} \left( \int_{-\infty}^{+\infty} |x(t)|^2 (1+t^2) \cdot dt \right)^{1/2}$$

$$\leq \left[ \text{Arctan} \Big|_{-\infty}^{\infty} \right]^{1/2} \cdot K$$

$$= \left( \frac{\pi}{2} - \left( -\frac{\pi}{2} \right) \right)^{1/2} \cdot K.$$

$$\boxed{= \sqrt{\pi} K < \infty}.$$



$$(iii) \quad x(t) = \begin{cases} \frac{1}{t^{2/3}} & 0 \leq t \leq 1 \\ 0 & t = 0 \text{ otherwise} \end{cases}$$

$$\int_0^1 |x(t)| \cdot dt = \int_0^1 \frac{1}{t^{2/3}} \cdot dt = 3 t^{1/3} \Big|_0^1 = 3 < \infty$$

$$\text{However, } \int |x(t)|^2 \cdot dt = \int_0^1 t^{-4/3} \cdot dt$$

$$= -3 t^{-1/3} \Big|_0^1 = \infty.$$

Thus  $x(t) \in L$ , but  $x(t) \notin L_2$ .

### Exercise 3.11 #

$$\begin{aligned} \text{Var}[\bar{X} + \bar{Y}] &= E[\bar{X} + \bar{Y} - E[\bar{X} + \bar{Y}]]^2 \\ &= E\left[\left((\bar{X} - E[\bar{X}]) + (\bar{Y} - E[\bar{Y}])\right)^2\right] \\ &= \text{var}(\bar{X}) + \text{var}(\bar{Y}) + 2E[(\bar{X} - E[\bar{X}]) (\bar{Y} - E[\bar{Y}])]. \\ &= \text{var}(\bar{X}) + \text{var}(\bar{Y}) + 2 \text{Cov}(\bar{X}, \bar{Y}) \\ &\leq \text{var}(\bar{X}) + \text{var}(\bar{Y}) + 2 \sqrt{\text{var}(\bar{X})} \sqrt{\text{var}(\bar{Y})} \\ &= \left(\sqrt{\text{var}(\bar{X})} + \sqrt{\text{var}(\bar{Y})}\right)^2. \end{aligned}$$



## Exercise 3.12 #

$$\begin{aligned}
 \text{(i)} \quad |\langle u, v \rangle| &= \left| \sum_{i=1}^{\infty} a_i b_i \right| \\
 &\leq \sum_{i=1}^{\infty} |a_i| |b_i| \\
 &\leq \sum_{i=1}^{\infty} \frac{1}{2} (|a_i|^2 + |b_i|^2) \\
 &= \frac{1}{2} \left( \sum_i |a_i|^2 + \sum_i |b_i|^2 \right) \\
 &= \frac{1}{2} (\|u\|_2^2 + \|v\|_2^2).
 \end{aligned}$$

$$|\langle \alpha u, \beta v \rangle| = |\alpha| |\beta| |\langle u, v \rangle|$$

$$\leq \frac{1}{2} (|\alpha|^2 \|u\|_2^2 + |\beta|^2 \|v\|_2^2).$$

$$\text{subs: } |\alpha| = \frac{1}{\|u\|} \quad \text{and} \quad |\beta| = \frac{1}{\|v\|}$$

$$= \frac{1}{2} (1+1)$$

$$\frac{1}{\|u\|_2 \|v\|_2} |\langle u, v \rangle| \leq 1$$

$$|\langle u, v \rangle| \leq \|u\|_2 \|v\|_2$$

$$(ii) \{a_i'\}_{i=1}^{\infty}, \{b_i'\}_{i=1}^{\infty}$$

$$a_i' = \begin{cases} a_i & , 1 \leq i \leq d \\ 0 & , \text{else} \end{cases}$$

$$b_i' = \begin{cases} b_i & , 1 \leq i \leq d \\ 0 & , \text{else} \end{cases}$$

$$\langle a, b \rangle = \sum_{i=1}^d a_i b_i^* \quad , \quad \langle a, a \rangle = \sum_{i=1}^d |a_i|^2 = \|a\|^2$$

$$\text{since } \|a - tb\|^2 \geq 0 = \langle a - tb, a - tb \rangle \\ = \|a\|^2 - t \langle b, a \rangle - t^* \langle a, b \rangle + |t|^2 \|b\|^2$$

$$t = \frac{\langle b, a \rangle}{\|b\|^2} \quad , \quad (\|b\|^2 \neq 0).$$

$$0 \leq \|a\|^2 - \frac{|\langle a, b \rangle|^2}{\|b\|^2}$$

$$|\langle a, b \rangle|^2 \leq \|a\|^2 \|b\|^2$$

### Exercise 13.3 #

Treating matrices as vectors.

$$\langle A, B \rangle = \text{tr}(AB^T) = \sum_{k=1}^n \sum_{l=1}^m a^{(k,l)} b^{(k,l)}$$



$$\|A\|_F^2 = \langle A, A \rangle = \sum_{k=1}^n \sum_{l=1}^m (a^{(k,l)})^2$$

$$(\text{tr}(AB^T))^2 = \langle A, B \rangle^2 \leq \langle A, A \rangle \langle B, B \rangle = \left( \sum_{k,l} (a^{(k,l)})^2 \right) \left( \sum_{k,l} (b^{(k,l)})^2 \right)$$

Exercise 3.14#

$$\sum_i |a_i|^2 = \sum_i |a_i| |a_i|$$

$$= \sum_i \frac{|a_i|}{\sqrt{1+|a_i|}} \cdot |a_i| (\sqrt{1+|a_i|})$$

$$\leq \left[ \sum_i \frac{|a_i|^2}{(1+|a_i|)} \right]^{1/2} \left[ \sum_i |a_i|^2 (1+|a_i|) \right]^{1/2}$$

Now  $\sum |a_i| < \infty \Rightarrow \lim |a_i| = 0 \Rightarrow k$  such that  $|a_i| < k \quad \forall i \in \mathbb{N}$ .

$$\text{Thus, } = \left( \sum_{i=1}^{\infty} \frac{|a_i| |a_i|}{1+|a_i|} \right)^{1/2} \left( \sum_{i=1}^{\infty} |a_i|^2 (1+|a_i|) \right)^{1/2}$$

$$\leq \left( \sum_i |a_i| \right)^{1/2} \left( \sum_i |a_i|^2 (1+k) \right)^{1/2}$$

$$\text{Thus, } \left( \sum_{i=1}^{\infty} |a_i|^2 \right)^2 \leq \sum_{i=1}^{\infty} |a_i| (1+k) \sum_{i=1}^{\infty} |a_i|^2$$

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$$\sum_{i=1}^{\infty} |a_i| \leq \sum_{i=1}^{\infty} |a_i| (1+k) < \infty$$

Exercise 3.15 #

$$\underbrace{a_1 + a_2 + \dots + a_d}_{d} = \sum_{i=1}^d a_i \cdot \underbrace{1}_{d}$$

$$\leq \sqrt{\sum_{i=1}^d a_i^2} \sqrt{\sum_{i=1}^d \frac{1}{d^2}}$$

$$= \sqrt{a_1^2 + a_2^2 + \dots + a_d^2}$$

$$\sqrt{\frac{d^2}{d^2}}$$

$$= \sqrt{\frac{a_1^2 + a_2^2 + \dots + a_d^2}{d}}$$